Table 1: Key Units

N_{av}	$6.022\,14 \times 10^{23}\,\text{mol}^{-1}$
R	$8.314\,472\,\text{J}\,\text{mol}^{-1}\,\text{K}^{-1}$
R	$8.2057 \times 10^{-2}\,\text{L}\,\text{mol}^{-1}\,\text{K}^{-1}$
k_B	$1.380\,65 \times 10^{-23}\,\text{J}\,\text{K}^{-1}$
k_B	$8.617\,34 \times 10^{-5}\,\text{eV}\,\text{K}^{-1}$
h	$6.626\,07 \times 10^{-34}\,\text{J}\,\text{s}$
k_B/h	$20\,836\,619\,123.3\,\text{K}^{-1}\,\text{s}^{-1}$
c	$2.997\,924\,58 \times 10^{8}\,\text{m}\,\text{s}^{-1}$

Table 2: Stoichiometry Equations

	one reaction	reaction network
balanced reaction	$\sum_j \nu_j A_j = 0$	$\sum_j \nu_{ij} A_j = 0, \forall i$
advancement	$n_j = n_{j0} + \nu_j \xi$	$n_j = n_{j0} + \sum_i \nu_{ij} \xi_i$
in flow	$F_j = F_{j0} + \nu_j \dot{\xi}$	$F_j = F_{j0} + \sum_i \nu_{ij} \dot{\xi}_i$
mole fraction	$y_j = n_j / \sum_j n_j = F_j / \sum_j F_j$	
partial pressure	$P_j =_j P$	
concentration	$C_j = n_j/V = F_j/v$	
ideal gas law	$V = (RT/P) \sum_j n_j$	
in flow	$v = (RT/P) \sum_j F_j$	

Table 3: Activity Models

pure substance	$a_j = 1$
ideal gas	$a_j = y_j P/P^\circ$
solute	$a_j = C_j/C^\circ$

Table 4: Thermodynamic Equations

First Law	$dU = 0$ in any process in a closed system
Second Law	$dS \geq 0$ in any spontaneous process in a closed system
Third Law	$S(0\,\mathrm{K} = 0$
Enthalpy	$H = U + PV$
Gibbs energy	$G = H - TS$ minimized in any spontaneous reaction
std reaction enthalpy	$\Delta H^\circ_{\mathrm{rxn}}(T) = \sum_j \nu_j \Delta H^\circ_{f,j}(T)$
std reaction entropy	$\Delta S^\circ_{\mathrm{rxn}}(T) = \sum_j \nu_j S^\circ_j(T)$
mixture free energy	$G(n_j, T, P) = \sum_j n_j \mu_j$
reaction free energy gradient	$dG = \sum_j \mu_j dn_j$
chemical potential	$\mu_j = \mu^\circ_j(T) + RT \ln a_j$
chemical equilibrium	$\partial G / \partial \xi = 0$
mass action expression	$Q(\xi) = \prod_j a_j^{\nu_j}$
equilibrium constant	$K_{eq}(T) = \exp(-\Delta G^\circ(T)/RT) = \prod_j a^{\nu}_{j,\mathrm{eq}j}$
Gibbs-Helholtz relation	$\partial(G/T)/\partial T = -H/T^2$
van't Hoff approximation	$K_{\mathrm{eq}}(T_2)/T_2 - K_{\mathrm{eq}}(T_1)/T_1 = -\Delta H^\circ(T)/R$

Table 5: Kinetics equations

Arrhenius expression	$\ln k = -E_a/RT$		
	Rate	Integrated rate	half-life
First-order reaction	$r = kC_A$	$C_A = C_{A0} e^{-kt}$	$t_{1/2} = \ln 2/k$
Second-order reaction	$r = kC_A^2$	$1/C_A = 1/C_{A0} - kt$	$t_{1/2} = 1/C_{A0}k$